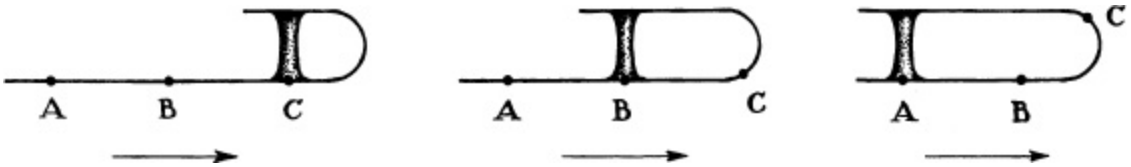
As an aid to understanding how the laws of physics control time's hookup, I thought about a more complicated situation. Suppose that one mouth of the wormhole is at rest in my living room and the other is in interstellar space, traveling away from Earth at nearly the speed of light. And suppose that, despite this relative motion of its two mouths, the wormhole's length (the length of its tunnel through hyperspace) remains always fixed at 30 centimeters. (Figure 14.6 explains how it is possible for the length of the wormhole to remain fixed while its mouths, as seen in the external Universe, move relative to each other.) Then, as seen in the external Universe, the two mouths are in different reference frames, frames that move at a high speed relative to each other; and *the mouths therefore must experience different flows of time.* On the other hand, as seen through the wormhole's interior, the mouths are at rest with respect to each other, so they share a common reference frame, which means that *the mouths must experience the same flow of time*. From the external viewpoint they experience different time flows, and from the internal viewpoint, the same time flow; how confusing!



14.6 Explanation of how the mouths of a wormhole can move relative to each other as seen in the external Universe, while the length of the wormhole remains fixed. Each of the diagrams is an embedding diagram like that in Figure 14.1, seen in profile. The diagrams are a sequence of snapshots that depict motion of the Universe and the wormhole *relative to hyperspace*. (Recall, however, that hyperspace is just a useful figment of our imaginations; there is no way that we as humans can ever see or experience it in reality; see Figures 3.2 and 3.3.) Relative to hyperspace, the bottom part of our Universe is sliding rightward in the diagrams, while the wormhole and the top part of our Universe remain at rest. Correspondingly, as seen in our Universe, the mouths of the wormhole are moving relative to each other (they are getting farther apart), but as seen through the wormhole they are at rest with respect to each other; the wormhole's length does not change.

Gradually, in my quiet isolation, the confusion subsided and all became clear. The laws of general relativity predict, unequivocally, the flow of time at the two mouths, and they predict, unequivocally, that the two time flows will be *the same* when compared through the wormhole, but will be *different* when compared outside the wormhole. Time, in this sense, hooks up to itself differently through the wormhole than through the external Universe, when the two mouths are moving relative to each other.